When perturbation theory is properly and effectively applied, the initial estimate will be reasonably close to the final answer; when incorporated, the fine details ignored in the initial estimate make small differences in the final result. But sometimes when you go to pay a final bill it is shockingly different from the initial estimate. Although you might use other, more emotive terms, technically this is called a failure of perturbation theory. This means that the initial approximation was not a good guide to the final answer because the "refinements," rather than causing relatively small deviations, resulted in large changes to the ballpark estimate.

As indicated briefly in earlier chapters, our discussion of string theory to this point has relied on a perturbative approach somewhat analogous to that used by the mechanic. The "incomplete understanding" of string theory that we have referred to from time to time has its roots, in one way or another, in this approximation method. Let's build up to an understanding of this important remark by discussing perturbation theory in a context that is less abstract than string theory but closer to its string theory application than the example of the mechanic.

A Classical Example of Perturbation Theory

Understanding the motion of the earth through the solar system provides a classic example of using a perturbative approach. On such large distance scales, we need consider only the gravitational force, but unless further approximations are made, the equations encountered are extremely complicated. Remember that according to both Newton and Einstein, everything exerts a gravitational influence on everything else, and this quickly leads to a complex and mathematically intractable gravitational tug-of-war involving the earth, the sun, the moon, the other planets, and, in principle, all other heavenly bodies as well. As you can imagine, it is impossible to take all of these influences into account and determine the exact motion of the earth. In fact, even if there were only three heavenly participants, the equations become so complicated that no one has been able to solve them in full.3

Nevertheless, we can predict the motion of the earth through the solar system with great accuracy by making use of a perturbative approach. The enormous mass of the sun, in comparison to that of every other member of our solar system, and its proximity to the earth,